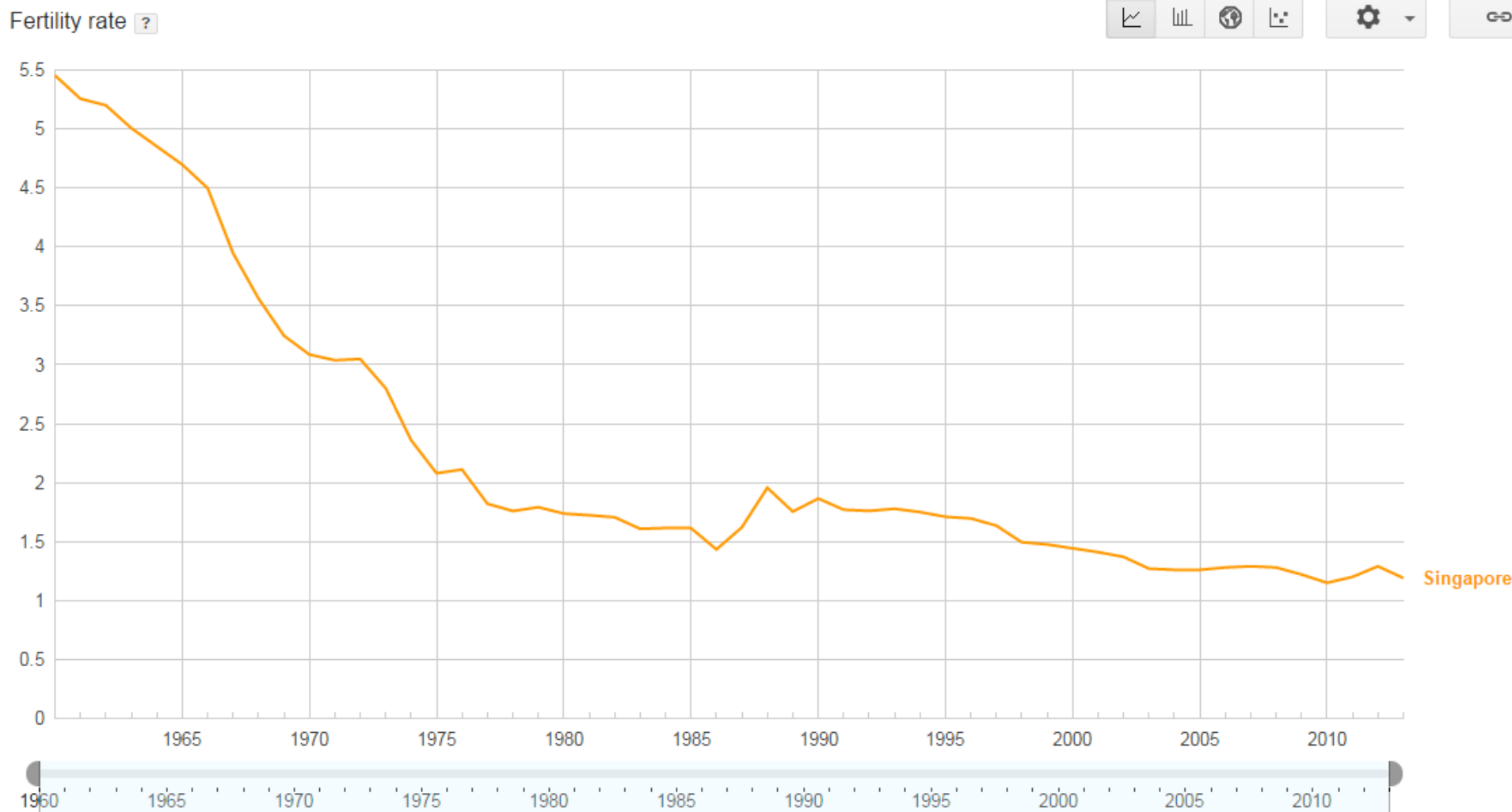


LECTURE 6: BINARY CHOICE MODELS

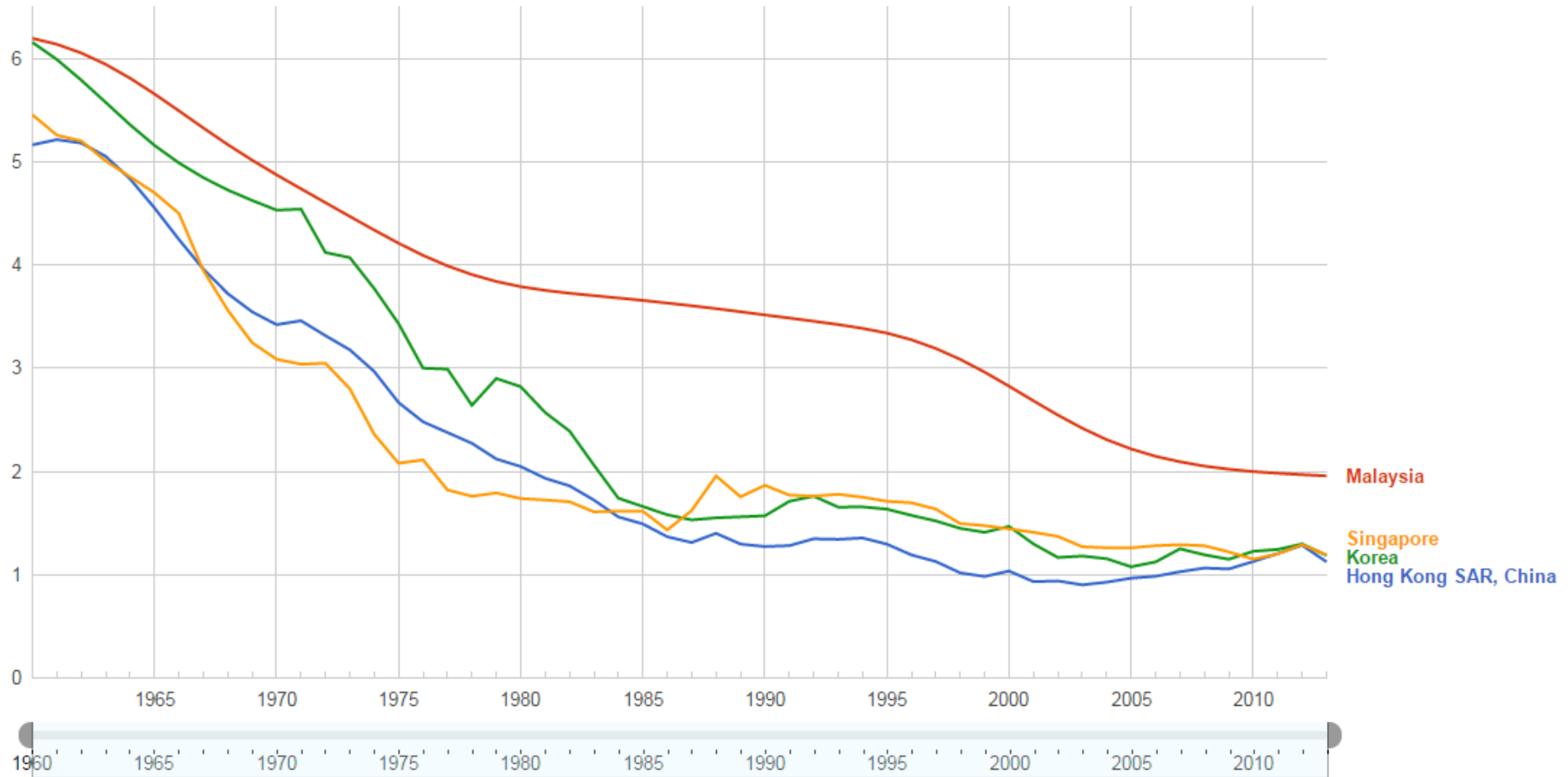
- (A) Introduction to Binary Decisions
- (B) Linear Probability Model (LPM)
- (C) Probit Model
- (D) Logit Model

(A) Introduction to Binary Decisions

Background 1: declining trend of total fertility rate in Singapore



Background 2: Similar pattern in South Korea, Hong Kong



- There are a few reasons why the fertility rate is declining in Singapore. For example, Living Cost, Marriage Age, Education Level, Bachelorism, etc. To women (or men), whether to have a baby or not is a yes or no decision, i.e., binary choices.
- In 2015, more Singaporean couples were ready to bear children. “*At least 33,793 Jubilee babies born in 2015, the highest in 13 years.*” The Straits Times (Feb 2, 2016)
- Why was 2015 so special?
Government subsidies (e.g. baby bonus scheme)
- We could evaluate the effect of “baby bonus” or other packages on fertility intention.
 - (i) Evaluate how this policy (or temporary change) affects the fertility rate.
 - (ii) Evaluate how this policy affects people’s binary decision. (microdata)

- There are many examples related to binary choices. We could study what the factors affect people's binary decisions and quantify the impacts of the factors.

Examples:

- female labor force participation: work or be a housewife
 - marriage decision: married or single
 - applying for mortgage: accepted or denied
 - go for graduate school: yes or no
 - support or not support (the government)
 - buy or not buy (a car)
- These decision variables involve YES or NO. Statistically, we use $y = 1$ and 0 to represent people's decisions.

(B) Linear Probability Model

- If the y_i is a binary dependent variable, then the linear regression model is called the linear probability model (LPM).

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} \dots + \beta_K x_{iK} + u_i, \quad i = 1, 2, \dots, n$$

Zero Conditional Mean assumption: $E(u_i | x_{i1}, \dots, x_{iK}) = 0$ implies

$$E(y_i | x_{i1}, \dots, x_{iK}) = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} \dots + \beta_K x_{iK}$$

- What is $E(y_i | x_{i1}, \dots, x_{iK})$?

Since y_i is a binary (dummy) variable,

$$E(y_i) = 1 \times \Pr(y_i = 1) + 0 \times \Pr(y_i = 0) = \Pr(y_i = 1).$$

If we conditional on the x variables, then the expression is as follows.

$$E(y_i | x_{i1}, \dots, x_{iK}) = \Pr(y_i = 1 | x_{i1}, \dots, x_{iK}) = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} \dots + \beta_K x_{iK}$$

- $E(y_i|x_{i1}, \dots, x_{iK})$, the population regression model, tells us the probability of $y_i = 1$ given the values of $x_1, x_2, \dots, x_K$.
- The β_j measures the partial effect of x_j on the probability of $y = 1$.

Because,

$$E[y|x_1, x_2, \dots, x_K] = Pr(y = 1|x_1, x_2, \dots, x_K) = \beta_0 + \beta_1 x_1 + \dots + \beta_k x_k$$

So,

$$\beta_j = \frac{\partial E[y|x_1, x_2, \dots, x_K]}{\partial x_j} = \frac{\partial Pr(y = 1|x_1, x_2, \dots, x_K)}{\partial x_j}, j = 1, \dots, k$$

- Just like running the usual regression model, we can easily obtain the estimated regression: $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \dots + \hat{\beta}_k x_k$

Example: To buy or not to buy a car?

There are many reasons why one buys (or not to buy) a car. For example, household's income, COE price, household size, home location, etc.

Consider the model $y_i = \beta_0 + \beta_1 COE_i + \beta_2 income_i + \beta_3 household\ size_i + u_i$, where

y_i : 1 if one has a car(s) and 0 otherwise,

COE_i : The COE price the person i needs to pay when s/he made the decision,

$income_i$: household's income (in \$1,000 unit) ,

$household\ size_i$: number of family members.

Q1: What is the expected sign for the β_2 slope coefficient?

Answer:

Q2: Suppose $\hat{\beta}_2 = 0.065$. How would you explain this estimated slope coefficient?

Answer:

- Married women's decision to participate in the labor force depends on many factors, e.g., husband's income, education, number of kids.

Example: $y_i = \beta_0 + \beta_1 nwifeinc_i + \beta_2 educ_i + \beta_3 kidslt6_i + u_i$,

where y_i : 1 for being in the labor force and 0 for being a housewife,

$nwifeinc_i$: husband's earnings,

$educ_i$: wife's education,

$kidslt6_i$: number of children less than 6 years old.

We use the Morz.dta dataset, run this regression model and obtain the result as follows.

```
. reg inlf nwifeinc educ kidslt6
```

Source	SS	df	MS	Number of obs =	753
Model	22.3586557	3	7.45288523	F(3, 749) =	34.38
Residual	162.3691	749	.216781175	Prob > F =	0.0000
				R-squared =	0.1210
				Adj R-squared =	0.1175
Total	184.727756	752	.245648611	Root MSE =	.4656

inlf	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
nwifeinc	-.0077404	.001519	-5.10	0.000	-.0107224	-.0047583
educ	.0572465	.0077912	7.35	0.000	.0419513	.0725418
kidslt6	-.2227047	.0325987	-6.83	0.000	-.2867004	-.158709
_cons	.0737593	.0931678	0.79	0.429	-.1091417	.2566604

Q: How do you interpret the coefficients in this regression?

Remarks:

(1) $\hat{y}_i$ can be smaller than 0 in this regression model if *nwifeinc* is sufficiently low or $\hat{y}_i$ can be greater than 1 if *educ* is sufficiently large.

(2) The dataset contains *husband's educ* and *location* variables. But they are not the factors that significantly affect women's decision.

(3) For the multiple regression $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_{i1} + \hat{\beta}_2 x_{i2} \dots + \hat{\beta}_K x_{iK}$, the ceteris paribus interpretation of $\hat{\beta}_1 = \frac{\partial \hat{y}}{\partial x_1}$, i.e., the change in predicted probability that $y_i = 1$ for a Δx_1 , given other factors held constant.

- Advantages of LPM:

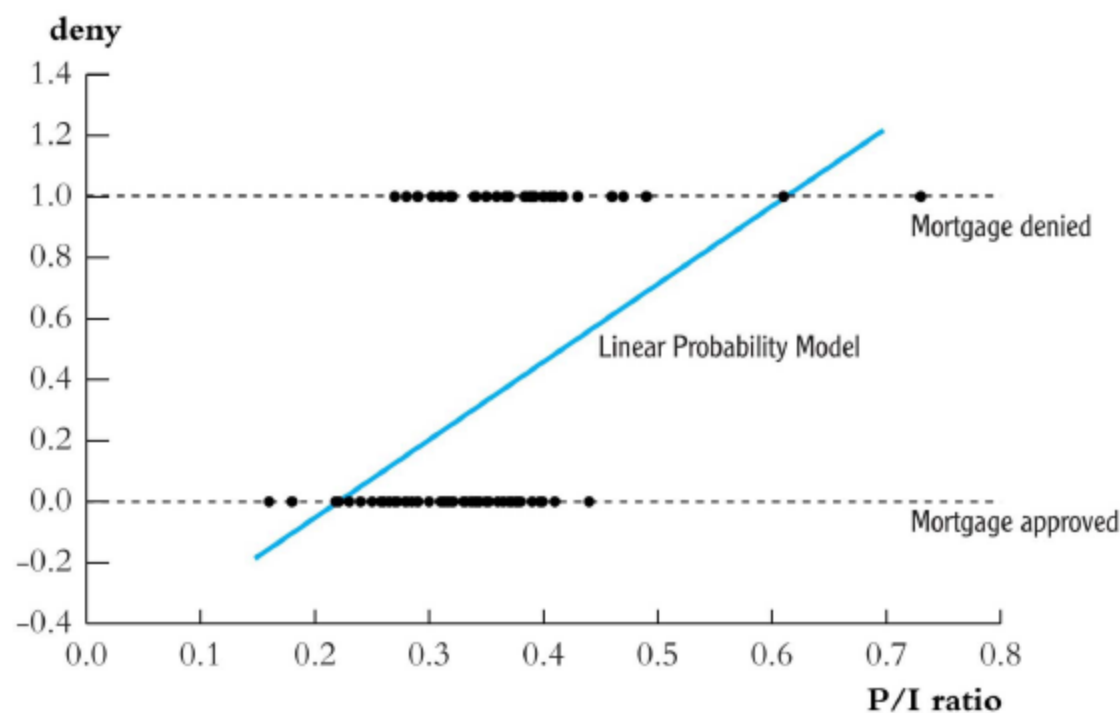
- Easy to implement (i.e., OLS) and simple to interpret results

- Straightforward to test hypothesis for example: $H_0: \beta_2 = 0$

- Disadvantage of LPM:
 - The predicted y_i (or the fitted probability) can be smaller than zero or greater than 1.
 - The predicted probability is linear in X. Therefore, the partial (or marginal) effect of any explanatory variable is constant. This doesn't make common sense. For instance, in the married women's decision to participate in the labor force example, the estimated coefficient for *educ* is 0.052, meaning for every additional year of education received, the probability of going to work constantly increases by 5.2% on average.

Example: Application to the Home Mortgage Disclosure ACT (HMDA) Data for Boston (Stock and Watson's textbook)

Y: Mortgage application denial rate, X: debt payments to income (P/I) ratio



Linear Probability Model: $deny = \beta_0 + \beta_1 \times P/I + u$, $n = 2380$

```
. reg deny piratio , robust
```

```
Linear regression      Number of obs   =      2,380
                      F(1, 2378)       =      37.56
                      Prob > F         =      0.0000
                      R-squared        =      0.0397
                      Root MSE      =      .31828
```

deny	Coef.	Robust Std. Err.	t	P> t	[95% Conf. Interval]	
piratio	.0060353	.0009848	6.13	0.000	.0041041	.0079666
_cons	-.0799096	.0319666	-2.50	0.012	-.1425949	-.0172243

Question: What is the predicted deny, $\widehat{deny}$ when P/I is 30%?

Answer: $Pr(deny = 1|P/I = 30) = \widehat{deny} = -0.08 + 0.00604 \times 30 = 0.1012$

Question: Calculate the effect of increasing P/I from 30% to 50%.

Answer: $Pr(deny = 1|PI_Ratio = 50) = \widehat{deny} = -0.08 + 0.00604 \times 50 = 0.222$

The probability of denial rate increases from 0.1012 to 0.222. The increase rate is more than 100%.

Linear Probability Model:

$deny = \beta_0 + \beta_1 \times \frac{P}{I} + \beta_2 black + u$, where deny and black are both dummy variables.

```
. reg deny piratio black , robust
```

```
Linear regression               Number of obs   =       2,380
                               F(2, 2377)         =       49.39
                               Prob > F           =       0.0000
                               R-squared           =       0.0760
                               Root MSE        =       .31228
```

deny	Coef.	Robust Std. Err.	t	P> t	[95% Conf. Interval]	
piratio	.0055919	.0008867	6.31	0.000	.0038532	.0073307
black	.1774282	.0249463	7.11	0.000	.1285096	.2263469
_cons	-.0905136	.0285996	-3.16	0.002	-.1465963	-.0344309

The coefficient on black is significant at 1%. At first glance, we would conclude that given P/I (in terms of %) being equal, the black people on average have a 17.7% higher probability to get mortgage applications denied compared to nonblack people.

The 17.7% estimate is surprising high. Can it really be the case? If not, what are the problems on this regression result?

Remarks:

- (a) The linear probability model (LPM) is a linear function of $x_1, x_2, \dots, x_K$.

$$E(y|x_1, \dots, x_K) = Pr(y = 1|x_1, \dots, x_K) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 \dots + \beta_K x_K$$

- (b) It is easy to estimate and interpret. To estimate the model, we *always* use heteroskedasticity-robust standard errors because the homoscedasticity assumption is violated in the LPM.
- (c) For the extreme cases, the predicted y_i can be smaller than zero or greater than 1.
- (d) The predicted probability is linear in X. Therefore, the partial (or marginal) effect of any explanatory variable is constant. This doesn't make common sense. For instance, in the women's decision to participate the labor force, the estimated coefficient for *educ* is 0.052, meaning for every additional year of education received, the probability of going to work constantly increases by 5.2% on average.
- (e) We could use the nonlinear models (probit and logit) to solve the problems mentioned in (c) and (d).

(C) Probit Model

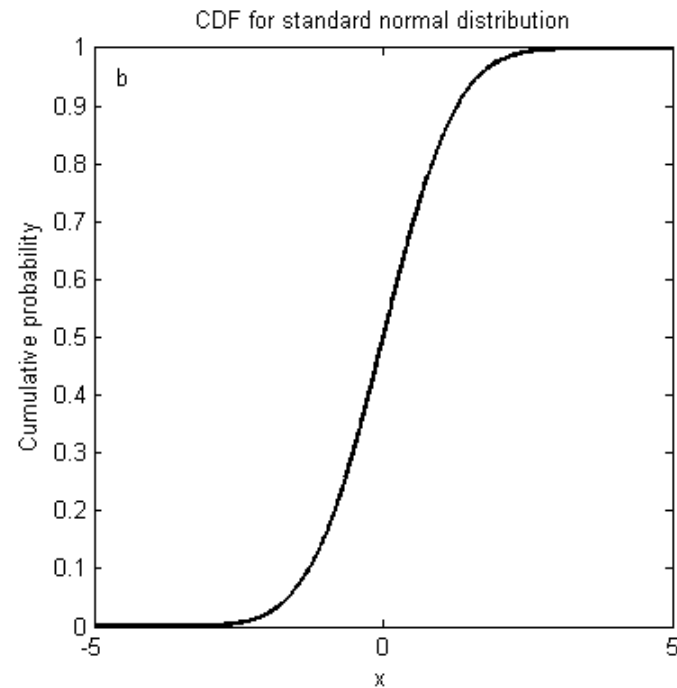
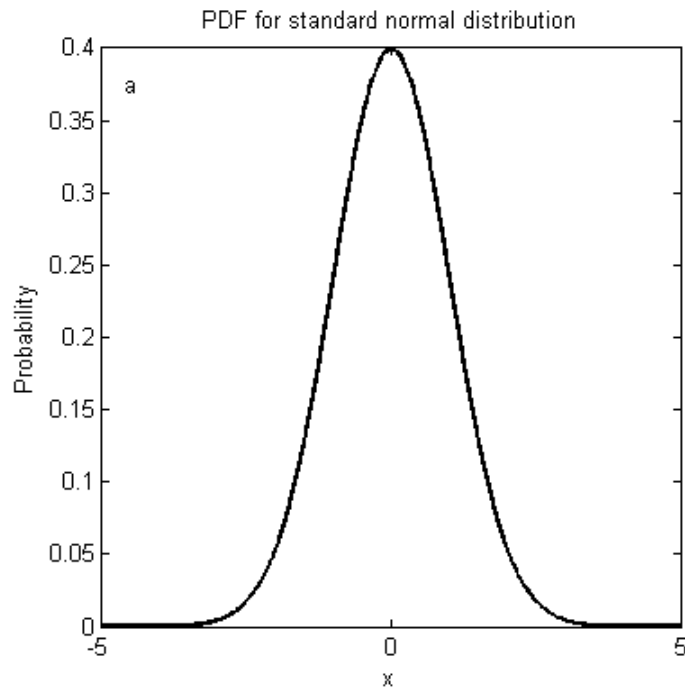
- To fix the LPM “problems”, we could consider a class of binary response models of the form $Pr(y = 1|x_1, \dots, x_K) = G(\beta_0 + \beta_1 x_1 + \beta_2 x_2 \dots + \beta_K x_K)$, where G is a function taking on values between zero and one: $0 \leq G(.) \leq 1$.
- Statistically, we can choose the cumulative standard normal density function as the G function, i.e. $G(z) = \Phi(X\beta)$. This is known as the probit model,

$$Pr(y = 1|x_1, \dots, x_K) = \Phi(\beta_0 + \beta_1 x_1 + \beta_2 x_2 \dots + \beta_K x_K).$$

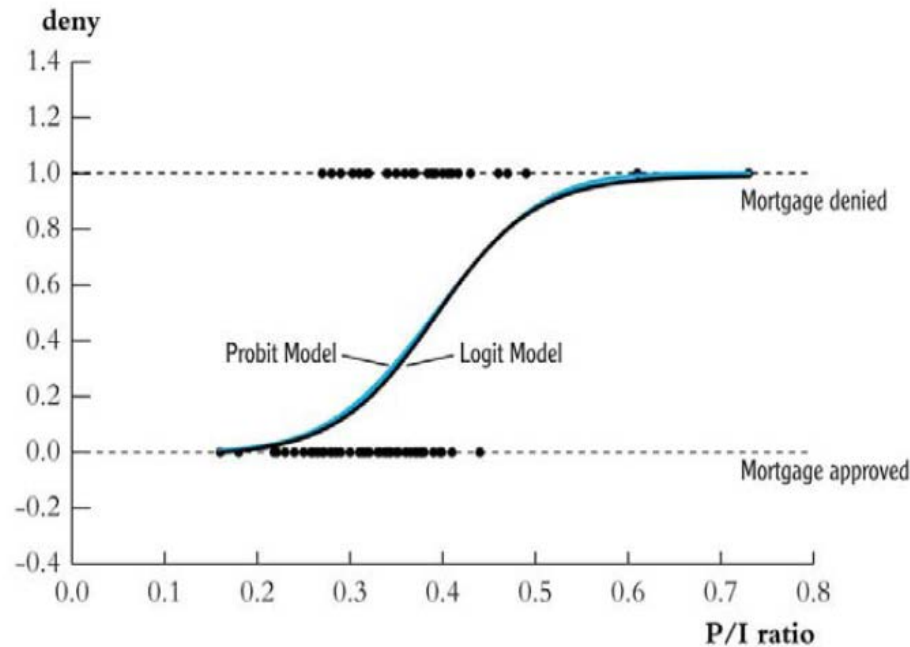
Given any combinations of real numbers $\mathbf{x} = (x_1, \dots, x_K)$, this function Φ produces probabilities in between 0 and 1, i.e., $\Phi(\mathbf{x}) \in [0,1]$. This CDF is symmetric, smooth and the 0 and 1 are asymptotes.

- The limitations of the LPM are overcome by using more sophisticated probit

- Review: HE1004/HE1005/ MH2500 levels



- $\Phi(x) = \int_{-\infty}^x \phi(v)dv$, where $\Phi(\cdot)$ is a standard normal CDF and $\phi(\cdot)$ is the PDF.
 - (a) $\phi(x) = (2\pi)^{-\frac{1}{2}} \times \exp\left(-\frac{x^2}{2}\right)$
 - (b) When $x \rightarrow -\infty$, $\Phi(x) \rightarrow 0$
 - (c) When $x \rightarrow \infty$, $\Phi(x) \rightarrow 1$



- For the probit model,

$$E[y|\mathbf{x}] = Pr(y = 1|\mathbf{x}) = G(\beta_0 + \beta_1 x_1 + \beta_2 x_2 \dots + \beta_K x_K) = \Phi(\mathbf{x}\beta).$$

A larger $\mathbf{x}\beta$ value corresponds to a larger $\Phi(\cdot)$ and that means the event ($y = 1$) is more likely to occur on average (as the probability increases).

- $\Phi(\mathbf{x}\beta) = \Phi(z) = \int_{-\infty}^z \phi(v)dv$, where $z = \mathbf{x}\beta$ and $\phi(\cdot)$ is a standard normal PDF.

➤ (a) $\phi(z) = (2\pi)^{-\frac{1}{2}} \times \exp\left(-\frac{z^2}{2}\right)$

➤ (b) When $z \rightarrow -\infty$, $\Phi(z) \rightarrow 0$

➤ (c) When $z \rightarrow \infty$, $\Phi(z) \rightarrow 1$

- For the probit model, the partial effect of x_j on $Pr(y = 1|x_1, \dots, x_K)$ is not constant and it is related to the values of $x_1, \dots, x_K$.

- $$\frac{\partial E[y|x_1, x_2, \dots, x_K]}{\partial x_j} = \frac{\partial Pr(y=1|x_1, x_2, \dots, x_K)}{\partial x_j}$$

$$= \phi(\beta_0 + \beta_1 x_1 + \beta_2 x_2 \dots + \beta_K x_K) \times \beta_j, \text{ where } \phi(z) = \frac{d\Phi(z)}{dz}$$

- Since $\phi(\beta_0 + \beta_1 x_1 + \beta_2 x_2 \dots + \beta_K x_K) > 0$, the partial effect always has the same sign as β_j . To get the partial effect numerical answer, people need to plug some interesting numbers to $x_1, x_2, \dots, x_K$ in the $\phi(\cdot)$ function.
- One popular method used in Stata is to replace $x_1, x_2, \dots, x_K$ variables with its sample average, i.e. $\phi(\hat{\beta}_0 + \hat{\beta}_1 \bar{x}_1 + \hat{\beta}_2 \bar{x}_2 \dots + \hat{\beta}_K \bar{x}_K)$.

$$\frac{\partial Pr(y = 1 | \bar{x}_1, \dots, \bar{x}_K)}{\partial x_j} = \phi(\hat{\beta}_0 + \hat{\beta}_1 \bar{x}_1 + \hat{\beta}_2 \bar{x}_2 \dots + \hat{\beta}_K \bar{x}_K) \times \hat{\beta}_j.$$

This is the partial effect of x_j for the “average” person in the sample or known as partial effect at the average.

➤ Potential problem : averages of them represent no one in the sample

Example: Married women's decision to participate in the labor force

Stata> probit inlf nwifeinc educ kidslt6

```
Iteration 0:    log likelihood = -514.8732
Iteration 1:    log likelihood = -465.52685
Iteration 2:    log likelihood = -465.45302
Iteration 3:    log likelihood = -465.45302
```

Probit regression

```
Number of obs   =      753
LR chi2(3)      =      98.84
Prob > chi2     =      0.0000
Pseudo R2      =      0.0960
```

Log likelihood = -465.45302

inlf	Coef.	Std. Err.	z	P> z	[95% Conf. Interval]	
nwifeinc	-.0231133	.0045451	-5.09	0.000	-.0320216	-.014205
educ	.1666664	.0235149	7.09	0.000	.120578	.2127547
kidslt6	-.6525247	.0996888	-6.55	0.000	-.8479112	-.4571382
_cons	-1.245253	.2714193	-4.59	0.000	-1.777225	-.7132805

- We use the MLE method. It's a numerical optimization problem. The iteration log shows fast convergence in 3 iterations. Empirically, a large number of iterations may signal a high degree of multicollinearity.
- For large samples, $\hat{\beta}_{MLE}$ estimators are consistent, normally distributed and asymptotically efficient among all the competing estimators (provided that the probit model is a correct model).
- The estimated model is

$$\hat{y}_i = \Phi(-1.24 - 0.023nwifeinc + 0.166educ - 0.652kidslt6)$$

(d) All the $\hat{\beta}_{MLE}$ signs are as expected and statistically significant different from 0.

For example:

$$\text{Recall: } \frac{\partial \Pr(y = 1 | \bar{x}_1, \dots, \bar{x}_K)}{\partial x_j} = \phi(\hat{\beta}_0 + \hat{\beta}_1 \bar{x}_1 + \hat{\beta}_2 \bar{x}_2 \dots + \hat{\beta}_K \bar{x}_K) \times \hat{\beta}_j$$

$$\begin{aligned} & \frac{\partial \Pr(y = 1 | \bar{x}_1, \dots, \bar{x}_K)}{\partial educ} \\ &= \phi(-1.24 - 0.023nwifeinc + 0.166educ - 0.652kidslt6) \times (0.166) > 0 \end{aligned}$$

Given other being equal, if *educ* increases, the probability of $y = 1$ increases.

✚ Suppose we want to know the partial effect of *educ* on $y = 1$ evaluated at means. We could plug the sample means of *nwifeinc*, *educ* and *kidslt6* into $\phi(\cdot)$.

Stata> mfx

Marginal effects after probit
 $y = \text{Pr}(\text{inlf})$ (predict)
 = .57228348

variable	dy/dx	Std. Err.	z	P> z	[	95% C.I.	]	X
nwifeinc	-.0090691	.00195	-4.66	0.000	-.012884	-.005254		20.129
educ	.0653958	.00925	7.07	0.000	.047259	.083532		12.2869
kidslt6	-.2560349	.03869	-6.62	0.000	-.331857	-.180213		.237716

(a) The predicted $\text{Pr}(y = 1|\bar{x}_1, \bar{x}_2, \bar{x}_3) = 0.572$ for this “average” person.

(b) If she takes one more year of *educ*, then the chance that she is working increases by 6.53%.

(c) If she has one more kid smaller than 6 years old, then the chance that she is working decreases by 25%.

Note: Let’s use (b) as an example. We could explain 0.065 as above because

$$\frac{\partial \widehat{\text{Pr}}(y = 1|\bar{x})}{\partial \text{educ}}$$

$$= \phi(-1.24 - 0.023 \times 20.12 + 0.166 \times 12.28 - 0.652 \times 0.23) \times 0.166 = 0.065$$

- Suppose we want to know the predicted probability for women who don't have any kids younger than 6 years old.

Stata> mfx, at (kidslt6=0) [partial effect evaluated at means except kidslt6 is evaluated at 0]

warning: no value assigned in at() for variables nwifeinc educ;
means used for nwifeinc educ

Marginal effects after probit
y = Pr(inlf) (predict)
= .63205694

variable	dy/dx	Std. Err.	z	P> z	[95% C.I.]	X
nwifeinc	-.0008711	.00187	-4.65	0.000	-.01238 -.005042	20.129
educ	.0628134	.00885	7.09	0.000	.045458 .080168	12.2869
kidslt6	-.245924	.03543	-6.94	0.000	-.315359 -.176489	0

$$\begin{aligned}
 Pr(y = 1 | \widehat{\bar{x}_1}, \bar{x}_2, x_3 = 0) \\
 &= \Phi(-1.24 - 0.023 \times nwifeinc + 0.166 \times educ + -0.652 \times kidslt6) \\
 &= \Phi(-1.24 - 0.023 \times 20.12 + 0.166 \times 12.28 - 0.652 \times 0) \\
 &= 0.623
 \end{aligned}$$

(D) Logit Model

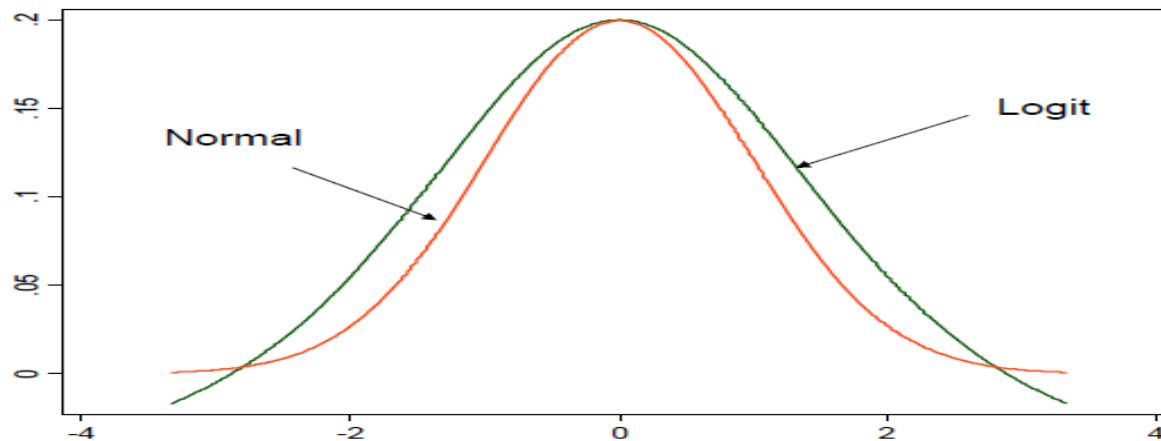
- If the $G(\cdot)$ function is a logistic function, then it is called the logit model.

$$\begin{aligned} Pr(y = 1|x_1, \dots, x_K) &= G(\beta_0 + \beta_1 x_1 + \beta_2 x_2 \dots + \beta_K x_K) \\ &= \frac{\exp(\beta_0 + \beta_1 x_1 + \beta_2 x_2 \dots + \beta_K x_K)}{1 + \exp(\beta_0 + \beta_1 x_1 + \beta_2 x_2 \dots + \beta_K x_K)}. \end{aligned}$$

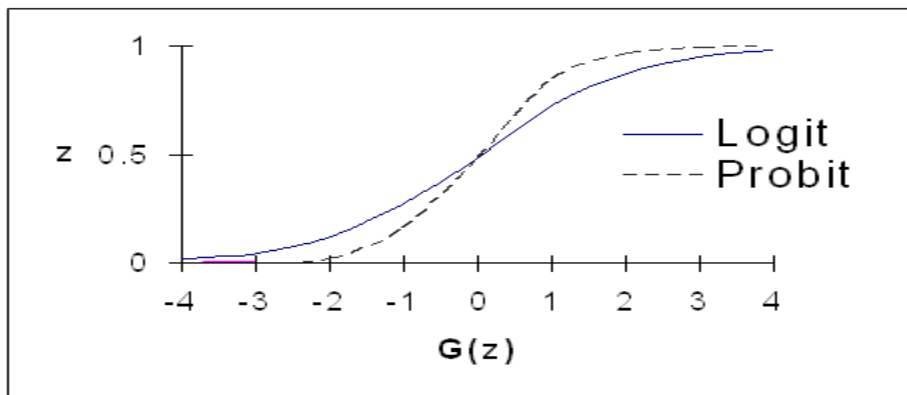
Given any combinations of values $\mathbf{x} = (x_1, \dots, x_K)$, this function always produces probabilities in between 0 and 1.

- The obvious difference between the logit and probit models is that the functional form $G(\cdot)$ used in the regression model. In the logit model, we use the logistic function.
- The logit model is estimated using ML. The estimators are normally distributed when $n \rightarrow \infty$.

- The graph is the PDFs for the standard normal (for probit) and logit distributions. Both have a similar distribution, but the normal distribution has fatter tails than logit distribution.



- The CDFs of the probit and logit functions are about linear in between $Pr(.) = 0.2$ and 0.8 . Therefore, for many cases LPM produces similar results as probit and logit models, except in the extreme values of $Pr(.)$. Additionally, probit reaches 0 and 1 quicker than logit.



Example:

$$\Pr(y = 1 | nwifeinc, educ, kidslt6) = G(\beta_0 + \beta_1 nwifeinc + \beta_2 educ + \beta_3 kidslt6)$$

Suppose the $G(\cdot)$ function is the cumulative standard logistic function instead of $\Phi(\cdot)$.

Stata> logit inlf nwifeinc educ kidslt6

```
Iteration 0:  log likelihood = -514.8732
Iteration 1:  log likelihood = -465.68562
Iteration 2:  log likelihood = -465.55378
Iteration 3:  log likelihood = -465.55373
Iteration 4:  log likelihood = -465.55373
```

Logistic regression

```
Number of obs   =      753
LR chi2(3)      =      98.64
Prob > chi2     =      0.0000
Pseudo R2      =      0.0958
```

Log likelihood = -465.55373

inlf	Coef.	Std. Err.	z	P> z	[95% Conf. Interval]	
nwifeinc	-.0385731	.0078654	-4.90	0.000	-.0539889	-.0231572
educ	.2741035	.0399979	6.85	0.000	.1957091	.3524979
kidslt6	-1.068074	.1671882	-6.39	0.000	-1.395757	-.7403916
_cons	-2.046709	.4552613	-4.50	0.000	-2.939005	-1.154413

- How to find out the partial effect of x_j on $Pr(y = 1|x_1, x_2, \dots, x_K)$?

$$\begin{aligned}
 \frac{\partial E[y|x_1, x_2, \dots, x_K]}{\partial x_j} &= \frac{\partial Pr(y = 1|x_1, x_2, \dots, x_K)}{\partial x_j} \\
 &= \frac{\partial G(\beta_0 + \beta_1 x_1 + \beta_2 x_2 \dots + \beta_K x_K)}{\partial x_j} \\
 &= \frac{\partial \left[\frac{\exp(\beta_0 + \beta_1 x_1 + \beta_2 x_2 \dots + \beta_K x_K)}{1 + \exp(\beta_0 + \beta_1 x_1 + \beta_2 x_2 \dots + \beta_K x_K)} \right]}{\partial x_j} \\
 &= \frac{\exp(\beta_0 + \beta_1 x_1 + \beta_2 x_2 \dots + \beta_K x_K)}{(1 + \exp(\beta_0 + \beta_1 x_1 + \beta_2 x_2 \dots + \beta_K x_K))^2} \times \beta_j
 \end{aligned}$$

[Note: HE1003 level]

. mfx

Marginal effects after logit
 y = Pr(inlf) (predict)
 = .57219848

variable	dy/dx	Std. Err.	z	P> z	[95% C.I.]	X
nwifeinc	-.0094422	.00193	-4.90	0.000	-.013222 -.005663	20.129
educ	.0670971	.00977	6.87	0.000	.047943 .086251	12.2869
kidslt6	-.2614511	.04111	-6.36	0.000	-.342023 -.180879	.237716

$$\begin{aligned}
 & \frac{\partial \Pr(y = 1|x)}{\partial \text{educ}} \\
 &= \frac{\exp(-2 - 0.03 \times \text{nwifeinc} + 0.274 \times \text{educ} - 1.07 \times \text{kidslt6})}{(1 + \exp(-2 - 0.03 \times \text{nwifeinc} + 0.274 \times \text{educ} - 1.07 \times \text{kidslt6}))^2} \times 0.274 \\
 &= \frac{\exp(-2 - 0.03 \times 20.12 + 0.274 \times 12.28 - 1.07 \times 0.23)}{(1 + \exp(-2 - 0.03 \times 20.12 + 0.274 \times 12.28 - 1.07 \times 0.23))^2} \times 0.274 \\
 &= 0.067
 \end{aligned}$$

- (a) The predicted $\Pr(y = 1|\bar{x}_1, \bar{x}_2, \bar{x}_3) = 0.572$ for this “average” person.
- (b) If she takes one more year of *educ*, then the chance that she is working increases by 6.7%. (The probit model gave us the answer 6.53%)
- (c) If she has one more kid smaller than 6 years old, then the chance that she is working decreases by 26%. (The probit model gave us the answer 25%)

Remarks:

1. The partial effects in the probit and logit models are not constant. They are related to the values of $x_1, x_2, \dots, x_K$. The slope parameter β_j is NOT the partial effect of x_j on the probability of “y=1”.

Therefore, the slope coefficients in the LPM, probit and logit models are different. We don't compare the slope coefficients from different models directly.

2 Hypothesis Testing in the probit and logit models: same as in OLS

Example 1: $H_0: \beta_1 = \beta_2 = \beta_3 = 0$ in the logit model

Stata commands:

```
logit inlf nwifeinc educ kidslt6
```

```
test nwifeinc educ kidslt6
```

```
. test nwifeinc educ kidslt6  
( 1) [inlf]nwifeinc = 0  
( 2) [inlf]educ = 0  
( 3) [inlf]kidslt6 = 0  
  
      chi2( 3) =    78.00  
Prob > chi2 =    0.0000
```

H_0 is rejected since the p -value is 0.

Example 2: linear restriction: $H_0: 2\beta_2 - \beta_3 = 0$

```
. test 2*educ- kidslt6=0
( 1) 2*[inlf]educ - [inlf]kidslt6 = 0
      chi2( 1) =    65.05
      Prob > chi2 =    0.0000
```

H_0 is rejected since the p -value is 0.